



FLAT (CSE 2221) MID TERM SCHEME

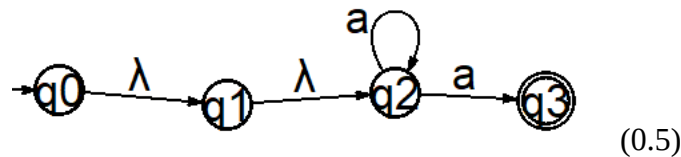
1. Which of the following is true about the language accepted by an NFA and its equivalent DFA?(0.5)

- a) They accept different languages.
- b) The NFA accepts a superset of the language accepted by the DFA.
- **c) They accept the same language.
- d) The DFA accepts a superset of the language accepted by the NFA.

2. The minimum number of states required in a DFA to accept a language $L = \{a^n b^{2m} \mid n, m \geq 1\}$ (0.5)

- a) n
- b) m
- c) 4
- d) ** 5

3. What is the minimum length of string to reach the final state in the below given NFA.



- a) 0
- b) **1
- c) 2
- d) 4

4. If $w = \text{abbab}$ the set of all prefixes of the string w is given as: (0.5)

- a. $\{a, ab, abb, abba, abbab\}$
- b. $\{a, ab, abb, abba, abbabbb\}$
- c. ** $\{\lambda, a, ab, abb, abba, abbab\}$
- d. $\{\lambda, a, ab, bb, abb, bba, abba, abbab\}$

5. Find the regular expression for all strings of size at least two over $\Sigma = \{0, 1\}$ (0.5)

1. ****(0+1)(0+1)(0+1)***

2. $(0+1)^*(0+1)^*$

3. $(0+1)^*(0+1)$

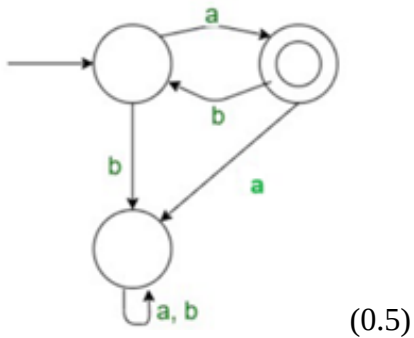
4. $(0+1)(0+1)$

5. Identify the non-regular grammar from the following: (0.5)

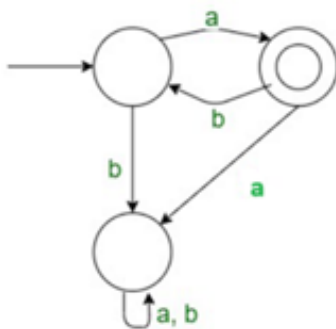
- 1. $S \rightarrow abS \mid a$
- 2. $S \rightarrow S1ab$, $S1 \rightarrow S1ab \mid S2$, $S2 \rightarrow a \mid \lambda$
- 3. $S \rightarrow abS$, $S \rightarrow bB \mid \lambda$, $B \rightarrow b$
- 4. ****** $S \rightarrow aAB \mid a$, $A \rightarrow Ba \mid b$, $B \rightarrow b$

6. Which of these will match the strings Rover, rover, and their plural variants only? (0.5)

1. $(r+R)\text{over}(s)^*$
2. $(R) \text{ rover } (s)$
3. $(r.R)\text{over}(s)^*$
4. $^{**}(r+R)\text{over}(s+\lambda)$
7. Find the regular expression for the given DFA.



1. aba^*
2. $^{**}(ab)^*a$
3. abb^*
4. aba
8. Find the regular expression for the given DFA.(0.5)



5. aba^*
6. $^{**}(ab)^*a$
7. abb^*
9. How many strings of length less than 4 contain the language described by the regular expression $(x+y)^*y(a+ab)^*$?
a) 7

- b) 10
- c) 11
- d) **12

10. Find the equivalent of the Regular Expression with $A = a^*$, $B = b^*$ for $AB = ?$

- a) $\{a^n b^n \mid n \geq 0\}$
- b) $\{a^m b^n \mid m \geq 0, n \geq 0\}$
- c) $(a+b)^*$
- d) $(ab)^*$

11. Using the pumping lemma to probe that the language $L = \{0^k 1^k : k \geq 0\}$ is not regular. (3)

Solution: (each step carries .5 marks)

If a language L is a regular language, then there exists a pumping length $p \geq 1$ such that any string w in L with $|w| \geq p$ can be divided into three parts: $w = xyz$

Where:

1. $|xy| \leq p$ (the length of the first two parts is at most p).
2. $|y| \geq 1$ (the middle part must not be empty).
3. For all $i \geq 0$, the string $xy^i z$ must also be in L (you can "pump" the y part).

Proof by Contradiction:

1. We will assume that $L = \{0^k 1^k : k \geq 0\}$ is **regular**, and show that this leads to a contradiction using the pumping lemma.
2. Let the pumping length be p and choose a string (w) from the language.

Let $w = 0^p 1^p$ where p is the pumping length given by the lemma.

3. By the lemma, we can write: $w = xyz$

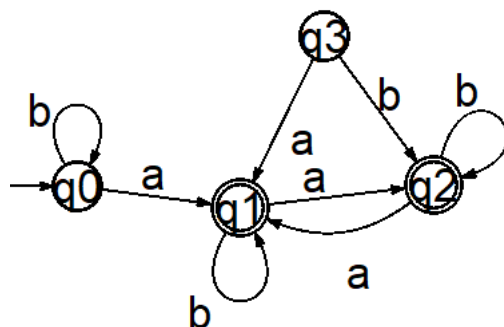
Where:

$|xy| \leq p$ and $|y| \geq 1$

Let $x = 0^a$, $y = 0^b$ and $z = 0^{p-a-b} 1^p$

4. Let $i = 2$ then the new pumped string $w' = xy^2z$ will be of the form $0^{p+b} 1^p$ which has more 0's than 1's.
5. But all strings in L must have **equal numbers of 0's and 1's**.
6. Since $w' \notin L$, this **contradicts** the assumption that L is a regular language.

12. Minimize the states of the DFA given below using mark and reduce method. (4)



Solution :

1. Removal of state q3 as it not reachable.

½ marks

2. Transition table after removal of q3

½ marks

δ	a	b
q0	q1	q0
q1	q2	q1
q2	q1	q2

3. Filling the table with (final and no-final or no-final and final combination of states)

1mark

	q0	q1
q1	✓	
q2	✓	

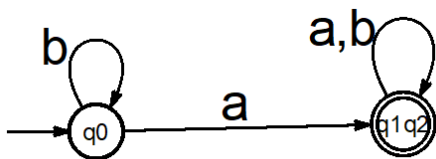
4. Decide to mark the unmarked pair (q1,q2) on input a and b 1mark

Using $\delta(q1,a) \rightarrow q2$ and $\delta(q2,a) \rightarrow q1$ as q1 ,q2) unmarked check with b input

5. $\delta(q1,b) \rightarrow q1$ and $\delta(q2,b) \rightarrow q2$ as (q1 ,q2) unmarked b further marking of (q1,q2) mis not done.

6. Merge the states q1q2 and draw the DFA with initial and final state.

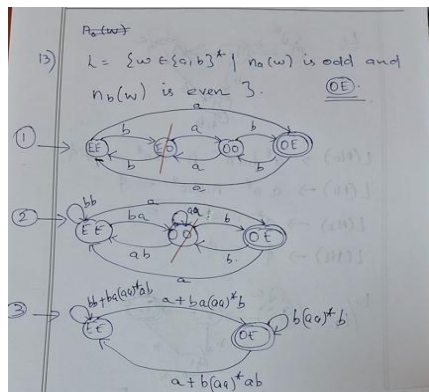
1marke



13. Find the regular expression for the language $L = \{w \in \{a, b\}^* : n_a(w) \text{ is odd and } n_b(w) \text{ is even}\}$ using GTG method. (4)

solution:

(2+2)



Ans.

$$r_1 = bb + ba(aa)^*ab$$

$$r_2 = a + ba(aa)^*b$$

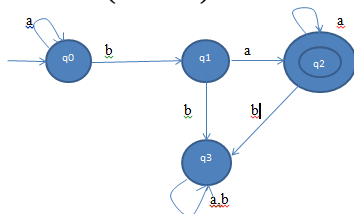
$$r_3 = a + b(aa)^*ab$$

$$r_4 = b(aa)^*b$$

Sub $r_1^* r_2 (r_4 + r_3 r_2)^*$

14. Find L_1/L_2 by constructing a DFA for the language $L_1 = L(a^*baa^*)$ and modify the DFA to accept L_1/L_2 for the language $L_2 = L(aba^*)$. (3)

solution: (1+1+1)=3



$L_2 = L(aba^*)$

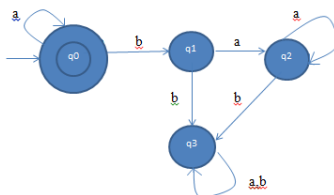
$$L(M_0) \cap L(aba^*) = L(a^*baa^*) \cap L(aba^*) \neq \emptyset$$

$$L(M_1) \cap L(aba^*) = L(aa^*) \cap L(aba^*) = \emptyset$$

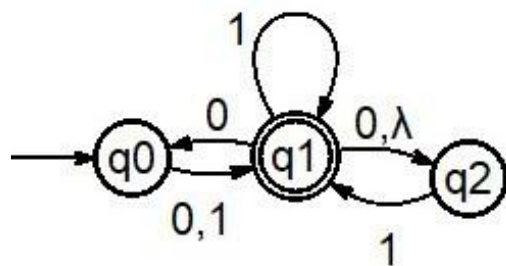
$$L(M_2) \cap L(aba^*) = L(a^*) \cap L(aba^*) = \emptyset$$

$$L(M_3) \cap L(aba^*) = \emptyset \cap L(aba^*) = \emptyset$$

$L_1/L_2 = L(a^*)$



15. Convert the NFA given below to its equivalent DFA.(4)



Solution: (1+1+1)=3

δ	0	1
q0	q1,q2	q1
q1,q2	q0,q2	q1
q1	q0,q2	q1

Transition table for the DFA:-

δ	0	1
$\rightarrow q_0$	$\{q_1, q_2\}$	$\{q_1, q_2\}$
$\{q_1, q_2\}$	$\{q_0, q_2\}$	$\{q_1, q_2\}$
$\{q_0, q_2\}$	$\{q_1, q_2\}$	$\{q_1, q_2\}$

The states are: q_0 , $\{q_1, q_2\}$ (Final), $\{q_0, q_2\}$



16. Find the grammar for the Language $L = \{ w : |w| \bmod 3 > 0 \}$ on $\Sigma = \{a\}$.(3)

Solution: $G = (\{A, B, S\}, \{a\}, S, P)$ where P is as given below:

$S \rightarrow aA$ 1M
 $A \rightarrow aB \mid \lambda$ 1M
 $B \rightarrow aS \mid \lambda$ 1Ms

Or

$S \rightarrow aaaA \mid A$
 $A \rightarrow a$
 $A \rightarrow aa$

17. Give the right linear grammar to generate a language $L = \{ 00^*(01+0)^* \}$ on $\Sigma = \{0,1\}$.(2)

Solution: $G = (\{A, S\}, \{a\}, S, P)$ where P is given as $S \rightarrow 0A$; $A \rightarrow OA \mid 01A \mid \lambda$ (4X.5=2)

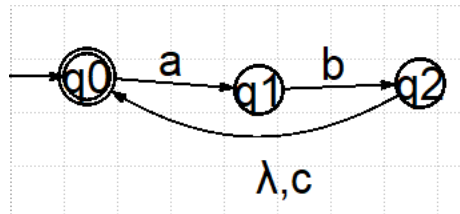
18. Construct a regular expression for the set of all strings over $\{0, 1\}$ that do not contain the substring "101".(2)

Solution: $RE = (0+100+11)(0+100+11)^*(001+\lambda)$

(2marks)

Q19. Construct an NFA with three states to accept the language $L = (ab, abc)^*$ on $\Sigma = \{a, b, c\}$. (2)

Solution:



2m